

What is Diabetes?

A simple, honest explanation of what is happening in your body, and why it matters.

You have probably heard the word diabetes a hundred times. Maybe from a doctor, a family member, or a news headline. But most people, even those who have it, do not fully understand what is actually happening inside their body when diabetes takes hold.

That is not your fault. For decades, diabetes education has been written in complicated medical language, buried in pamphlets nobody reads, or rushed through in a ten-minute doctor's appointment. You deserved better than that.

So here it is, explained clearly, honestly, and in plain English.

you will learn:

- What diabetes actually is and why it happens
- The difference between Type 1, Type 2, and Prediabetes
- How your body uses sugar and what goes wrong
- What A1C is and what your numbers mean
- What happens if it goes unmanaged, and what you can do about it

how your body is supposed to work.

To understand diabetes, you need to understand one simple process. No medical degree required.

Your body runs on glucose.

Every bite of food you eat, whether rice, bread, fruit, or even vegetables, gets broken down into a simple sugar called glucose. This glucose travels through your bloodstream, and its job is to fuel everything: your muscles, your brain, your organs. It is your body's main source of energy.

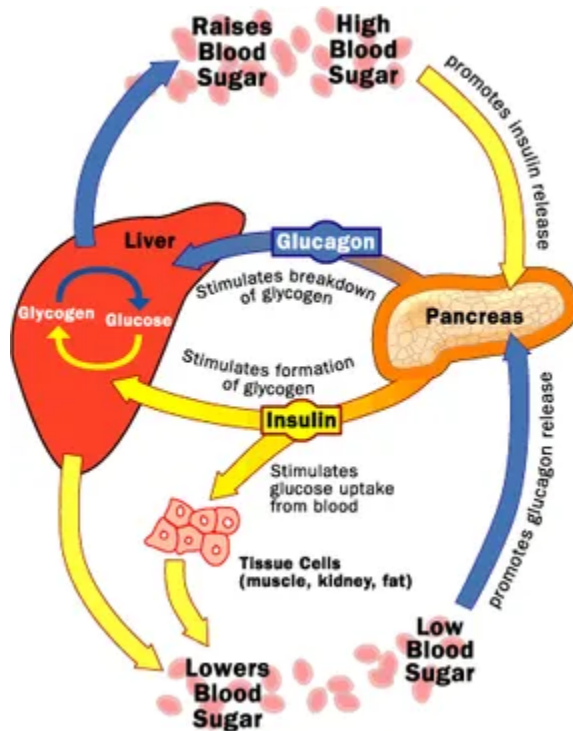
But glucose cannot just float into your cells on its own. It needs a key to get in. That key is a hormone called insulin.

Insulin is the key.

Insulin is made by an organ called the pancreas, which sits behind your stomach. The moment your blood sugar rises after a meal, your pancreas detects it and releases insulin. The insulin travels through your blood and physically unlocks your cells, letting glucose enter and be converted into energy.

Once glucose enters the cells, your blood sugar drops back to normal. Any extra glucose that is not needed right away gets stored in your liver for later. The system resets. Everything is balanced.

(we can do a animated diagram how glucose was used) ex:



So what goes wrong in diabetes?

In simple terms, the key stops working, or disappears entirely. Depending on the type of diabetes, it breaks down in different ways. But the result is the same: glucose cannot get into your cells properly, and it builds up in your blood.

That buildup of glucose floating in the bloodstream for too long is what causes all the damage. Not tomorrow. Not next week. Quietly, over months and years, high blood sugar slowly wears away at your blood vessels, nerves, and organs.

The three main types you need to know.

Type 1 Diabetes: The immune system attacks.

In Type 1, your own immune system makes a devastating mistake. It attacks and destroys the beta cells in your pancreas, the only cells that produce insulin. No beta cells means no insulin. No insulin means glucose cannot enter cells at all.

Type 1 is not caused by diet, lifestyle, or anything you did wrong. It is an autoimmune condition, meaning your body's defense system turned against itself. It can develop at any age, though it is most commonly diagnosed in children and young adults.

People with Type 1 diabetes must take insulin every single day to survive.

Key fact: Type 1 accounts for about 5 to 10 percent of all diabetes cases. It is not preventable, and it is not caused by eating too much sugar.

Type 2 Diabetes: The key stops working.

Type 2 is the most common form, accounting for about 90 to 95 percent of all diabetes cases worldwide. Here, the pancreas still makes insulin. The problem is that your cells have stopped responding to it properly. This is called insulin resistance.

Think of it this way: the key exists, but the lock is jammed. Your pancreas tries to compensate by making more and more insulin, working overtime to force the door open. For a while, it manages. But eventually the pancreas gets exhausted and cannot keep up. Blood sugar rises. And stays high.

Type 2 is strongly connected to lifestyle. Diet, exercise, weight, and sleep all play a role. But genetics matter too. Many people develop Type 2 without any obvious lifestyle causes. And critically, it often causes no symptoms for years. That is why so many people, like my dad, only find out when the damage is already serious.

Important: Unlike Type 1, Type 2 diabetes can in many cases be managed, or even reversed, through lifestyle changes. You have more power here than you may think.

Prediabetes: The warning window.

Prediabetes is when your blood sugar is higher than normal, but not yet high enough to be classified as Type 2 diabetes. Think of it as your body's warning signal, like a flashing light saying something needs to change.

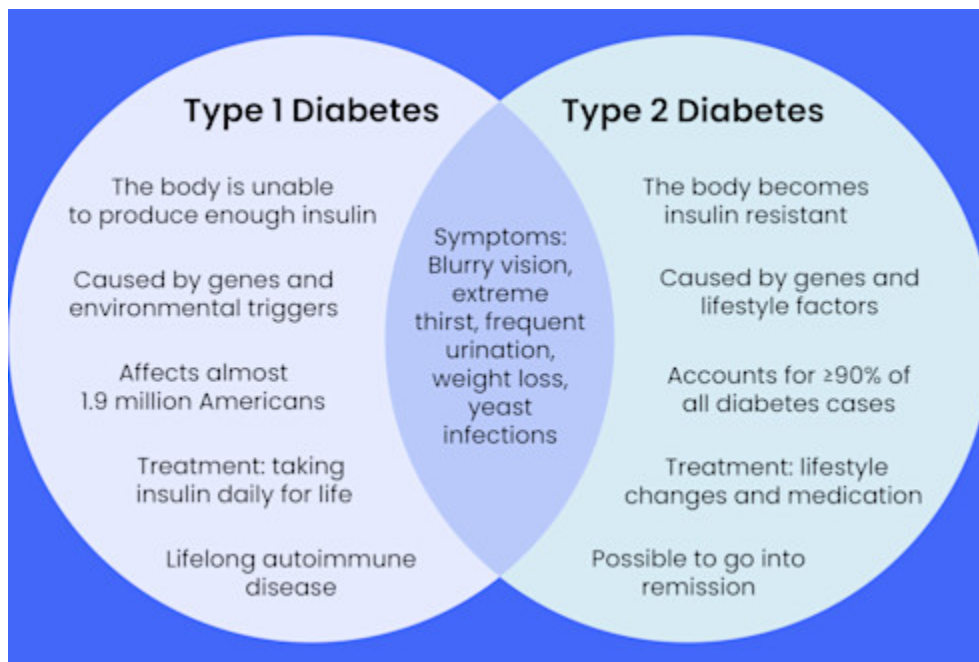
This is the most important stage to catch. With the right changes in diet and exercise at this point, many people avoid ever developing Type 2 diabetes at all. The window is open, and it can stay open for years.

Did you know: About 96 million American adults have prediabetes, and more than 80 percent of them do not know it. Prediabetes rarely has noticeable symptoms.



Type 1 vs. Type 2 vs. Prediabetes, side by side.

(visual note: a table is need here for comparison)ex:



(i forgot the one for pre-diabetic, we will decide later.)

What causes it?

Type 1: The immune system destroys insulin-making cells. Not caused by lifestyle.

Type 2: Cells stop responding to insulin, known as insulin resistance.

Prediabetes: Early insulin resistance. Blood sugar is rising but not yet in the diabetic range.

Can you prevent it?

Type 1: No. It is autoimmune and not related to lifestyle choices.

Type 2: Often yes. Diet, exercise, and weight management help greatly.

Prediabetes: Yes. Lifestyle changes are highly effective at this stage.

Does it go away?

Type 1: No. It is a lifelong condition.

Type 2: Not a permanent cure, but remission is possible through lifestyle changes.

Prediabetes: Reversible. Many people return to normal blood sugar fully.

Does it need insulin?

Type 1: Always. A person with Type 1 cannot survive without insulin.

Type 2: Sometimes, depending on how far it has progressed and how it is managed.

Prediabetes: No. Managed through lifestyle changes first.

Understanding your numbers.

When your doctor talks about your blood sugar or diabetes, they will throw around a lot of numbers. Here is what they actually mean.

What is A1C?

The A1C test, also called HbA1c, is the most important test for understanding your diabetes. It measures your average blood sugar level over the past two to three months, not just a snapshot from one moment.

Glucose sticks to a protein in your red blood cells called hemoglobin. Red blood cells live for about three months. So by measuring how much glucose is stuck to them, doctors can see your average blood sugar over the last three months, even if you fasted before the test. The pizza you

had last Tuesday is in there. (we can create a illustration or some sort of visual notes here): ex:

A1C and Estimated Average Glucose Levels

	A1C Percentage	Estimated Average Glucose (EAG)	
In-range	< 5.7%	< 117 mg/dL	6.5 mmol/L
Prediabetes	5.7-6.4%	117-137 mg/dL	6.5-7.6 mmol/L
Diabetes	> 6.4%	> 137 mg/dL	> 7.6 mmol/L
Increased risk of complications	6.5%	140 mg/dL	7.8 mmol/L
	7.0%	154 mg/dL	8.6 mmol/L
	7.5%	169 mg/dL	9.4 mmol/L
	8.0%	183 mg/dL	10.1 mmol/L
	8.5%	197 mg/dL	10.9 mmol/L
	9.0%	212 mg/dL	11.8 mmol/L
	9.5%	226 mg/dL	12.6 mmol/L
	10%	240 mg/dL	13.4 mmol/L

Below 5.7 percent: Normal. No diabetes.

5.7 to 6.4 percent: Prediabetes. This is the warning zone. Action now can prevent diabetes.

6.5 to 7.9 percent: Diabetes. Blood sugar needs to be managed.

8.0 percent and above: Poorly controlled diabetes. High risk of complications.

For most people with diabetes, the goal is to keep A1C below 7 percent. If yours is higher, that is not a reason to panic. It is a reason to act. Every 1 percent drop in A1C significantly reduces your risk of complications.

Daily blood sugar targets.

These are general targets. Always follow your doctor's specific range since individual goals vary based on age and health history.

Before meals (fasting): 80 to 130 mg/dL for people with diabetes. Below 100 mg/dL for people without diabetes.

One to two hours after eating: Below 180 mg/dL for people with diabetes. Below 140 mg/dL for people without diabetes.

At bedtime: 100 to 140 mg/dL for people with diabetes.

Why diabetes hides for so long.

One of the most dangerous things about Type 2 diabetes is that it often causes no noticeable symptoms, especially in the early years. You can have high blood sugar for a long time without feeling anything dramatic.

When symptoms do appear, they are easy to explain away.

(Visual note here, or any sort of illustration) (with pictures for comparison) examples ive found:



DHARMA
Homoeopathy

Feeling tired all the time: Most people say they have just been stressed and not sleeping enough.

Drinking water constantly: Most people say it has just been a hot week.

Blurry or changing vision: Most people say they probably need new glasses.

Needing to urinate more often: Most people say they just drink a lot of water.

Slow-healing cuts or bruises: Most people say they have always been a slow healer.

Tingling or numbness in feet: Most people say they have been sitting at their desk too long.

Frequent infections: Most people say they must have a weak immune system this season.

By the time symptoms feel impossible to ignore (severe fatigue, wounds that will not heal, vision problems, chest pain), the condition has often been building quietly for years. This is why so many people, including my dad, only discover their diabetes during an unrelated hospital visit, or when a routine blood test catches it almost by accident.

That is the power of getting tested, and staying tested. Early detection changes everything.

By the time my dad walked into the hospital, his diabetes had been there for years. He had no idea. Now you do.

What happens if diabetes goes unmanaged.

This section is not here to scare you. It is here because knowing the risks is what motivates real change. High blood sugar, over time, is like sticky syrup running through your pipes. It coats and damages the delicate blood vessels that deliver oxygen and nutrients to every organ in your body.

The good news is that most of these complications are preventable. With well-managed blood sugar, many people with diabetes live long, healthy lives with no serious complications at all. But catching and managing it early is what makes that possible.

(visual note here: illustration of people having severe diabetes, like rotted foot and black neck



and arms)

Your Eyes

High blood sugar damages the tiny blood vessels in the retina. Over time this causes diabetic retinopathy, which leads to blurry vision and, if left unmanaged, possible blindness. Annual eye exams can catch it early before serious damage occurs.

Your Kidneys

Your kidneys filter your blood all day long. Sustained high blood sugar damages their filtering units over time. The result can be chronic kidney disease, and in severe cases, dialysis or a transplant. The damage often shows no symptoms until it is advanced, which is why regular urine and blood tests matter.

Your Nerves

Nerve damage from diabetes is called neuropathy. It causes tingling, numbness, or burning pain, most often in the hands and feet. Because damaged nerves reduce your ability to feel pain, small cuts or sores can go unnoticed and become serious infections. Foot care becomes especially important.

Your Heart

People with diabetes are about twice as likely to develop heart disease compared to people without it. High blood sugar slowly stiffens and narrows blood vessels over time, raising the risk of heart attack and stroke. Managing blood sugar, blood pressure, and cholesterol together is the best protection.

You understand it now. That is everything.

When my dad was diagnosed, no one sat him down and explained any of this. He had to piece it together himself through late nights of researching, reading study after study, learning from other people who had been in his shoes.

He made changes. Hard ones. He changed what he ate, how he moved, how he slept. He tracked his numbers. He was patient with himself.

And he reversed it.

Not because he was special. But because he understood his condition well enough to fight it. That understanding, which you now have, is the first step.

What to do next:

- Talk to your doctor about getting an A1C test if you have not had one recently
- Write down any symptoms you have noticed, even small ones
- Read the next section: Why Did I Get It?
- Try the Blood Sugar Log tool to start tracking your numbers

Knowledge is not enough on its own. But it is where everything starts.

Sources

All content on this page is based on the following trusted medical sources.

- World Health Organization (WHO)- Diabetes Fact Sheet, 2024
- American Diabetes Association (ADA) - Standards of Care in Diabetes, 2024
- Centers for Disease Control and Prevention (CDC) - About Diabetes
- Cleveland Clinic -- Diabetes: What It Is, Types, Causes and Symptoms
- National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK)
- Joslin Diabetes Center - The Difference Between Type 1 and Type 2